

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

Fludarabine Kabi 50mg Lyophilized Powder for Solution for Injection or Infusion

DESCRIPTION:

Fludarabine Kabi 50mg Lyophilized Powder for Solution for Injection or Infusion, is a sterile lyophilized white or almost white powder or cake. The reconstituted solution is clear and colourless.

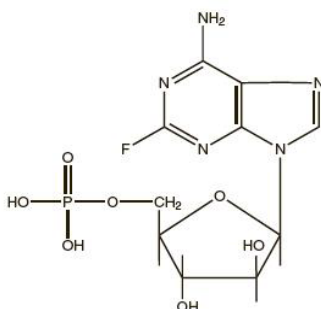
COMPOSITION:

Each Vial contains:

Fludarabine Phosphate	50 mg
Mannitol	50mg
Sodium Hydroxide	q.s. to adjust pH

CHEMICAL STRUCTURE:

The chemical name for Fludarabine phosphate is 9H-Purin-6-amine, 2-fluoro-9-(5-O-phosphono-β-D-arabinofuranosyl)(2-fluoro-ara-AMP). The molecular formula of Fludarabine phosphate is C₁₀H₁₃FN₅O₇P (molecular weight- 365.2) and the structure is:



PHARMACOLOGY:

Pharmacotherapeutic group: Antineoplastic agents, antimetabolites, purine analogues

ATC code: L01B B05

Fludarabine is a water-soluble fluorinated nucleotide analogue of the antiviral agent vidarabine, 9 β D arabinofuranosyladenine (ara A) that is relatively resistant to deamination by adenosine deaminase.

Fludarabine is rapidly dephosphorylated to 2F-ara-A which is taken up by cells and then phosphorylated intracellularly by deoxycytidine kinase to the active triphosphate, 2F ara ATP. This metabolite has been shown to inhibit ribonucleotide reductase, DNA (deoxyribonucleotide) polymerase α/δ and ε, DNA primase and DNA ligase thereby inhibiting DNA synthesis.

Furthermore, partial inhibition of RNA polymerase II and consequent reduction in protein synthesis occur.

While some aspects of the mechanism of action of 2F ara ATP are as yet unclear, it is assumed that effects on DNA, RNA (ribonucleic acid) and protein synthesis all contribute to inhibition of cell growth with inhibition of DNA synthesis being the dominant factor. In addition, in vitro studies have shown that exposure of CLL lymphocytes to 2F ara A triggers extensive DNA fragmentation and cell death characteristic of apoptosis.

Pharmacokinetics

- **Plasma and urinary pharmacokinetics of Fludarabine (2F-ara-A)**

The pharmacokinetics of fludarabine (2F ara A) have been studied after intravenous administration by rapid bolus injection and short-term infusion as well as following continuous infusion of fludarabine (2F ara AMP) in patients with malignant diseases.

2F ara AMP is a water soluble prodrug, which is rapidly and quantitatively dephosphorylated in the human organism to the nucleoside fludarabine (2F ara A). After single dose infusion of 25 mg 2F ara AMP per m² to cancer patients for 30 minutes 2F ara-A reached mean maximum concentrations in the plasma of 3.5 µM to 3.7 µM at the end of the infusion. Corresponding 2F-ara A levels after the fifth dose showed a moderate accumulation with mean maximum levels of 4.4 µM to 4.8 µM at the end of the infusion. During a 5 day treatment schedule (2F ara A) plasma trough levels increased by a factor of about 2. An accumulation of 2F-ara-A over several treatment cycles can be excluded. Post maximum levels decayed in three disposition phases with an initial half-life of approx. 5 minutes, an intermediate half-life of 12 hours and a terminal half life of approx. 20 hours.

An interstudy comparison of fludarabine pharmacokinetics resulted in a mean total plasma clearance (CL) of 79 ± 40 ml/min/m² (2.2 ± 1.2 ml/min/kg) and a mean volume of distribution (V_{ss}) of 83 ± 55 l/m² (2.4 ± 1.6 l/kg). Plasma levels of (2F ara A) and areas under the plasma level time curves increased linearly with the dose, whereas half-lives, plasma clearance and volumes of distribution remained constant independent of dose indicating a dose linear behaviour.

Occurrence of neutropenia and haematocrit changes indicated that a cytotoxicity of fludarabine depresses the haematopoiesis in a dose dependent manner.

2F ara A elimination is largely by renal excretion. 40% to 60% of the administered i.v. dose was excreted in the urine. Mass balance studies in laboratory animals with 3H 2F-ara AMP showed a complete recovery of radio-labelled substances in the urine. Another metabolite, 2F ara hypoxanthine, which represents the major metabolite in the dog, was observed in humans only to a minor extent. Individuals with impaired renal function exhibit a reduced total body clearance, indicating the need for a dose reduction. In vitro investigations with human plasma proteins revealed no pronounced tendency of 2F ara A protein binding.

- **Cellular pharmacokinetics of fludarabine triphosphate**

Fludarabine is actively transported into leukaemic cells, whereupon it is rephosphorylated to the monophosphate and subsequently to the di and triphosphate. The triphosphate 2F ara ATP is the major intracellular metabolite and the only metabolite known to have cytotoxic activity. Maximum 2F ara ATP levels in leukaemic lymphocytes of CLL patients were observed at a median of 4 hours and exhibited a considerable variation with a median peak concentration of approx. 20 µM.

2F ara ATP levels in leukaemic cells were always considerably higher than maximum 2F araA levels in the plasma indicating an accumulation at the target sites. In vitro incubation of leukaemic lymphocytes showed a linear relationship between extracellular 2F ara A exposure (product of 2F ara A concentration and duration of incubation) and intracellular 2F ara ATP enrichment. 2F ara ATP elimination from target cells showed median half-life values of 15 and 23 hours. No clear correlation was found between 2F ara A pharmacokinetics and treatment efficacy in cancer patients.

The total body clearance of the principle plasma metabolite 2F ara A shows a correlation with creatinine clearance, indicating the importance of the renal excretion pathway for the elimination of the compound. Patients with reduced renal function demonstrated an increased total body exposure (AUC of 2F ara A).

INDICATIONS:

Treatment of B cell chronic lymphocytic leukaemia (CLL) in patients with sufficient bone marrow reserves.

First line treatment with fludarabine should only be initiated in patients with advanced disease, Rai stages III/IV (binet stage C), or Rai stages I/II (Binet stage A/B) where the patient has disease related symptoms or evidence of progressive disease.

CONTRAINDICATIONS:

Hypersensitivity to the active substance or any of the excipients

Renal impairment with creatinine clearance < 30 ml/min.

Decompensated haemolytic anaemia.

Lactation.

ADVERSE EFFECTS:

Based on the experience with the use of fludarabine, the most common adverse events include myelosuppression (neutropenia, thrombocytopenia and anaemia), infection including pneumonia, cough, fever, fatigue, weakness, nausea, vomiting and diarrhoea. Other commonly reported events include chills, oedema, malaise, peripheral neuropathy, visual disturbance, anorexia, mucositis, stomatitis and skin rashes. Serious opportunistic infections have occurred in patients treated with fludarabine. Fatalities as a consequence of serious adverse events have been reported.

The table below reports adverse events by MedDRA system organ classes (MedDRA SOCs).

Table : Adverse events due to Fludarabine use

Organ Class/System	Very Common	Common	Uncommon	Rare	Not Known
Infections And infestations	Infections / Opportunistic infections (like latent viral reactivation, e.g. Progressive multifocal leucoencephalopathy, Herpes zoster virus, Epstein-Barr-virus), Pneumonia			Lymphoproliferative disorder (EBV associated)	
Neoplasms benign, malignant and unspecified (including cysts and		Myelodysplastic syndrome and Acute myeloid leukaemia (mainly associated with			

polyps)		prior, concomitant or subsequent treatment with alkylating agents, topoisomerase inhibitors or irradiation)			
Blood and lymphatic system disorders	Neutropenia, Anaemia, Thrombocytopenia	Myelosuppression			
Immune system disorders			Autoimmune disorder (including Autoimmune haemolytic anaemia, Evans syndrome, Thrombocytopenic purpura, Acquired haemophilia, Pemphigus)		
Metabolism and nutrition disorders		Anorexia	Tumour lysis syndrome (including Renal failure, Metabolic acidosis, Hyperkalaemia, Hypocalcemia, Hyperuricemia, Haematuria, Urate crystalluria, Hyperphosphatemia)		
Nervous system disorders		Neuropathy peripheral	Confusion	Coma, Seizures, Agitation	Cerebral haemorrhage
Eye disorders		Visual disturbance		Blindness, Optic neuritis, Optic neuropathy	
Cardiac disorders				Heart failure, Arrhythmia	
Respiratory, thoracic and mediastinal disorders	Cough		Pulmonary toxicity (including Pulmonary fibrosis, Pneumonitis,		Pulmonary haemorrhage

			Dyspnoea)		
Gastrointestinal disorders	Vomiting, Diarrhoea, Nausea	Stomatitis	Gastrointestinal haemorrhage, Pancreatic enzymes abnormal		
Hepatobiliary disorders			Hepatic enzyme abnormal		
Skin and subcutaneous tissue disorders		Rash		Skin cancer, Necrolysis epidermal toxic (Lyell type), Stevens-Johnson syndrome	
Renal and urinary disorder					Haemorrhagic cystitis
General disorders and administration site conditions	Fever, Fatigue, Weakness,	Oedema, Mucositis, Chills, Malaise			

Effects on ability to drive and use machines:

Fludarabine may reduce the ability to drive and use machines, since e.g. fatigue, weakness, visual disturbances, confusion, agitation and seizures have been observed.

DRUG INTERACTIONS:

The use of Fludarabine 50 mg Lyophilized Powder for Injection in combination with pentostatin is not recommended due to the risk of severe pulmonary toxicity.

The therapeutic efficacy of fludarabine phosphate may be reduced by dipyridamole and other inhibitors of adenosine uptake. Clinical studies and in vitro experiments have shown that during use of fludarabine in combination with cytarabine the intracellular peak concentration and intracellular exposure of Ara-CTP (active metabolite of cytarabine) increased in leukemic cells. Plasma concentrations of Ara-C and the elimination rate of Ara-CTP were not affected.

Incompatibilities

In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products except those mentioned in section for storage condition.

SPECIAL WARNINGS AND PRECAUTIONS FOR USE:

Myelosuppression

Severe bone marrow suppression, notably anaemia, thrombocytopenia and neutropenia, has been reported in patients treated with fludarabine.

Cumulative myelosuppression may be seen. While chemotherapy-induced myelosuppression is often reversible, administration of fludarabine requires careful haematological monitoring.

Fludarabine is a potent antineoplastic agent with potentially significant toxic side effects. Patients undergoing therapy should be closely observed for signs of haematologic and non-haematologic toxicity. Periodic assessment of peripheral blood counts is recommended to detect the development of anaemia, neutropenia and thrombocytopenia.

Several instances of trilineage bone marrow hypoplasia or aplasia resulting in pancytopenia, sometimes resulting in death, have been reported in adult patients. The duration of clinically significant cytopenia in the reported cases has ranged from approximately 2 months to approximately 1 year. These episodes have occurred both in previously treated or untreated patients.

As with other cytotoxics, caution must be exercised with fludarabine, when further haematopoietic stem cell sampling is considered.

Autoimmune disorders

Irrespective of any previous history of autoimmune processes or Coombs test status, life-threatening and sometimes fatal autoimmune phenomena have been reported to occur during or after treatment with fludarabine. The majority of patients experiencing haemolytic anaemia developed a recurrence in the haemolytic process after rechallenge with fludarabine.

Patients treated with fludarabine must be closely monitored for haemolysis. Discontinuation of therapy with fludarabine is recommended in case of haemolysis. Blood transfusion (irradiated) and adrenocorticoid preparations are the most common treatment measures for autoimmune haemolytic anaemia.

Neurotoxicity

The effect of chronic administration of fludarabine on the central nervous system is unknown. However, patients tolerated the recommended dose in some studies for relatively long term treatment times, (for up to 26 courses of therapy). Patients should be closely observed for signs of neurological side effects.

Tumour lysis syndrome

Tumour lysis syndrome associated with fludarabine treatment has been reported in CLL patients with large tumour burdens. Since fludarabine can induce a response as early as the first week of treatment precautions must be taken in those patients at risk of developing this complication.

Transfusion associated graft-versus-host disease

Transfusion-associated graft-versus-host disease (reaction by the transfused immunocompetent lymphocytes to the host) has been observed after transfusion of non-irradiated blood in fludarabine treated patients. Fatal outcome as a consequence of this disease has been reported with a high frequency. Therefore, to minimise the risk of transfusion-associated graft-versus-host disease, patients who require blood transfusion and who are undergoing, or have received treatment with fludarabine must receive irradiated blood only.

Skin cancer

The worsening or flare up of pre-existing skin cancer lesions as well as new onset of skin cancer has been reported in some patients to occur during or after fludarabine therapy.

Impaired state of health

In patients with impaired state of health, fludarabine must be given with caution and after careful risk/benefit consideration. This applies especially for patients with severe impairment of bone marrow function (thrombocytopenia, anaemia, and/or granulocytopenia), immunodeficiency or with a history of opportunistic infection.

Renal Impairment:

The total body clearance of the principal plasma metabolite 2-F-ara-A shows a correlation with creatinine clearance, indicating the importance of the renal excretion pathway for the elimination of the compound. Patients with reduced renal function demonstrated an increased total body exposure (AUC (area under curve) of 2F ara A). There are limited clinical data available in patients with impairment of renal function (creatinine clearance <70 ml/min).

Fludarabine must be administered cautiously in patients with renal impairment. In patients with moderate impairment of renal function (creatinine clearance between 30 ml/min and 70 ml/min), the dose should be reduced by up to 50% and the patient should be monitored closely). Fludarabine treatment is contraindicated if creatinine clearance is < 30 ml/min.

The elderly

Since there are limited data for the use of fludarabine in elderly persons (> 75 years), caution should be exercised with the administration in these patients. In patients aged 65 years or older, creatinine clearance should be measured before start of treatment.

Special precautions for disposal and other handling**Reconstitution**

Fludarabine must be prepared for parenteral use by aseptically adding sterile water for injections. When reconstituted with 2 ml of sterile water for injections, the powder should fully dissolve in less than 60 seconds. Each ml of the resulting solution will contain 25 mg of fludarabine phosphate, 25 mg of mannitol, and sodium hydroxide to adjust the pH to 7.7. The pH range for the final product is 7.2 - 8.2.

Dilution

The required dose (calculated on the basis of the patient's body surface) is drawn up into a syringe. For intravenous bolus injection this dose is further diluted into 10 ml of sodium chloride 9 mg/ml (0.9%). Alternatively, for infusion the required dose drawn up in a syringe may be diluted into 100 ml sodium chloride 9 mg/ml (0.9%) or 5% dextrose injection and infused over approximately 30 minutes.

The dilution compatibility study has been carried out in polyolefin infusion bags.

Inspection prior to use

Fludarabine must not be used in case of a defective container.

The reconstituted solution is clear and colourless. It should be visually inspected before use.

Only clear and colourless solutions without particles should be used.

Handling and disposal

Fludarabine must not be handled by pregnant staff. Procedures for proper handling should be followed according to local requirements for cytotoxic drugs. Caution should be exercised in the handling and preparation of the fludarabine solution. The use of latex gloves and safety glasses is recommended to avoid exposure in case of breakage of the vial or other accidental spillage. If the solution comes into contact with the skin or mucous membranes, the area should be washed thoroughly with soap and water. In the event of contact with the eyes, rinse them thoroughly with copious amounts of water. Exposure by inhalation should be avoided. The medicinal product is for single use only. Any unused product or waste material should be disposed of in accordance with local requirements.

Pregnancy

Fludarabine should not be used during pregnancy unless clearly necessary (e.g. life threatening situation, no alternative safer treatment available without compromising the therapeutic benefit,

treatment cannot be avoided). It has the potential to cause fetal harm. Prescribers may only consider the use of fludarabine, if the potential benefits justify the potential risks to the foetus.

Women should avoid becoming pregnant while on fludarabine therapy. Women of childbearing potential must be apprised of the potential hazard to the foetus.

Lactation

It is not known whether fludarabine or its metabolites are excreted in human milk. However, there is evidence from preclinical data that fludarabine and/or metabolites transfer from maternal blood to milk. Because of the potential for serious adverse reactions to fludarabine in breast fed infants. Fludarabine is contraindicated in nursing mothers.

Fertility

Women of childbearing potential must be apprised of the potential hazard to the foetus. Both sexually active men and women of childbearing potential must take effective contraceptive measures during and at least for 6 months after cessation of therapy.

Vaccination

During and after treatment with fludarabine, vaccination with live vaccines must be avoided.

Retreatment options after initial fludarabine treatment

A crossover from initial treatment with fludarabine to chlorambucil for non responders to fludarabine should be avoided because most patients who have been resistant to fludarabine have shown resistance to chlorambucil.

Excipients

Each vial of Fludarabine Kabi 50mg Lyophilized Powder for Solution for Injection or Infusion contains less than 1 mmol sodium (23 mg), i.e. essentially 'sodium-free'.

DOSAGE AND ADMINISTRATION:

Adults: The recommended dose is 25 mg fludarabine phosphate/m² body surface given daily for 5 consecutive days every 28 days by the intravenous route. Each vial is to be made up in 2 ml water for injection. Each ml of the resulting solution will contain 25 mg fludarabine phosphate (see also section Special Precautions for Disposal and Other Handling).

The required dose (calculated on the basis of the patient's body surface) of the reconstituted solution is drawn up into a syringe. For intravenous bolus injection this dose is further diluted into 10 ml of sodium chloride 9 mg/ml (0.9%). Alternatively, for infusion, the required dose drawn up in a syringe may be diluted into 100 ml sodium chloride 9 mg/ml (0.9%) and infused over approximately 30 minutes.

The duration of treatment depends on the treatment success and the tolerability of the drug. In CLL patients, Fludarabine 50mg Lyophilized Powder for Injection should be administered up to the achievement of best response (complete or partial remission, usually 6 cycles) and then the drug should be discontinued.

Special populations

Patients with hepatic impairment:

No data are available concerning the use of Fludarabine in patients with hepatic impairment. In this group of patients, Fludarabine 50mg Lyophilized Powder for Injection should be used with caution and administered if the perceived benefit outweighs any potential risk.

Patients with renal impairment:

Doses should be adjusted for patients with reduced kidney function. If creatinine clearance is between 30 and 70 ml/min, the dose should be reduced by up to 50% and close haematological monitoring should be used to assess toxicity (see section Special Warnings and Precautions for Use).

Fludarabine 50mg Lyophilized Powder for Injection treatment is contraindicated, if creatinine clearance is < 30 ml/min (see section Contraindications).

Children

The safety and effectiveness of Fludarabine in children has not been established. Therefore, Fludarabine 50mg Lyophilized Powder for Injection is not recommended for use in children.

Elderly patients

Since there are limited data for the use of fludarabine phosphate in elderly persons (> 75 years), caution should be exercised with the administration of fludarabine phosphate in these patients.

In patients over the age of 65 years, creatinine clearance should be measured, see "Patients with renal impairment" and section Special Warnings and Precautions for Use.

Method of Administration

Fludarabine 50 mg Lyophilized Powder for Injection should be administered under the supervision of a qualified physician experienced in the use of antineoplastic therapy. It is strongly recommended that Fludarabine 50mg Lyophilized Powder for Injection should be only administered intravenously. No cases have been reported in which paravenously administered Fludarabine 50 mg Lyophilized Powder for Injection led to severe local adverse reactions. However, unintentional paravenous administration must be avoided. For instructions on reconstitution of the medicinal product before administration, see section Special Precautions for Disposal and Other Handling.

OVERDOSE:

High doses of fludarabine have been associated with an irreversible central nervous system toxicity characterized by delayed blindness, coma, and death. High doses are also associated with severe thrombocytopenia and neutropenia due to bone marrow suppression. There is no known specific antidote for fludarabine overdosage. Treatment consists of drug discontinuation and supportive therapy.

STORAGE:

Store the vials in the original cartons at refrigerated temperature between 2°C to 8°C. After first opening, the product should be used immediately.

After reconstitution:

The physicochemical stability of the drug product after reconstitution in water for injections has been demonstrated for 24 hours at $25^{\circ}\text{C} \pm 2^{\circ}\text{C}$ / $60\% \pm 5\%\text{RH}$ and for 7 days at $5^{\circ}\text{C} \pm 3^{\circ}\text{C}$.

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2°C to 8°C .

After dilution:

The infusion solution is chemically stable when stored in infusion bags of 0.9 % sodium chloride solution or 5 % dextrose solution, for 48 hours when stored at 2°C to 8°C or 24 hours at 25°C under normal light conditions.

From a microbiological point of view however, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2°C to 8°C , unless dilution has taken place in a controlled and validated aseptic condition.

DOSAGE FORMS AND PACKAGING AVAILABLE:

Type-I, 6 ml, clear, moulded glass vials, closed with a 20 mm grey bromobutyl rubber stoppers and sealed with 20 mm green aluminum flip-off overseals.
Each pack contains 1 vial.

MANUFACTURER INFORMATION:

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